A Nonlinear approach to Viscoelasticity via Rational Extended Thermodynamics

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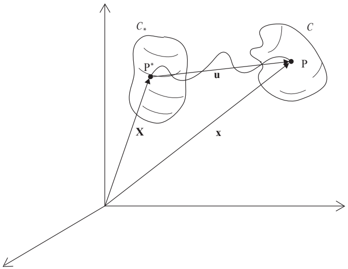
Continuum theory



A continuum can be identified with a region in Euclidean space. The vector function

$$\mathbf{x} \equiv \mathbf{x}(\mathbf{X}, t) \quad \Longleftrightarrow \quad x_i \equiv x_i(X_1, X_2, X_3, t), \quad i = 1, 2, 3, \tag{1}$$

provides a mapping between C^* and C. The *deformation gradient operator* $\mathbf{F}$ is defined as follows:

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \equiv (F_{iA}), \quad F_{iA} = \frac{\partial x_i}{\partial X_A}, \tag{2}$$

Balance Laws

A law is called a *balance law for a given density* $\Psi(\mathbf{x},t)$ if it can be represented as:

$$\frac{d}{dt}\int_V \Psi dV = -\int_\Sigma \Phi_i n_i d\Sigma + \int_V f dV \tag{3}$$

which, under regularity assumptions can be written locally in Eulerian variables as:

$$\frac{\partial \Psi}{\partial t} + \frac{\partial}{\partial x_i}\left(\Psi v_i + \Phi_i\right) = f \tag{4}$$

Or by introducing the quantities:

$$\Psi^* = \frac{\rho^*}{\rho}\Psi, \qquad \Phi_A^* = \Phi_i F_{iA}^C, \qquad f^* = \frac{\rho^*}{\rho} f \tag{5}$$

the same balance law in Lagrangian variables takes the form

$$\frac{\partial \Psi^*}{\partial t} + \frac{\partial \Phi_A^*}{\partial X_A} = f^* \tag{6}$$

Eulerian Variables:

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + \dfrac{\partial \rho v_i}{\partial x_i} = 0 \\[2ex] \dfrac{\partial \rho v_j}{\partial t} + \dfrac{\partial}{\partial x_i}\left(\rho v_i v_j - t_{ij}\right) = \rho b_j \qquad (j = 1,2,3) \\[2ex] \dfrac{\partial}{\partial t}\left(\frac{1}{2}\rho v^2 + \rho e\right) + \dfrac{\partial}{\partial x_i}\left\{\left(\frac{1}{2}\rho v^2 + \rho e\right) v_i - t_{ij} v_j + q_i\right\} = \rho b_j v_j + r. \end{cases} \tag{7}$$

Lagrangian Variables:

$$\begin{cases} \rho = \rho^*/J \\[2ex] \dfrac{\partial \rho^* v_j}{\partial t} - \dfrac{\partial T_{jA}}{\partial X_A} = \rho^* b_j \\[2ex] \dfrac{\partial}{\partial t}\left(\rho^* \dfrac{v^2}{2} + \rho^* e\right) + \dfrac{\partial}{\partial X_A}\left(Q_A - T_{iA} v_i\right) = \rho^* b_i v_i + r^*. \end{cases} \tag{8}$$

$$T_{jA} = t_{ij} F^C_{iA} \quad \Longleftrightarrow \quad \mathbf{T} = \mathbf{t}\mathbf{F}^C. \tag{9}$$

$$Q_A = q_i F^C_{iA} \quad \Longleftrightarrow \quad \mathbf{Q} = \mathbf{F}^{CT}\mathbf{q} \tag{10}$$

Constitutive equations

It is crucial to emphasize that in continuum theories, physical laws are expressed through "balance laws", necessitating "constitutive equations" to close the system. In the modern approach, these additional equations must adhere to universal principles like objectivity and compatibility with an entropy principle, representing the actual constitutive properties of the material.
Historically, constitutive equations were formulated mainly empirically, falling into three major classes:

- **Local constitutive equations** - Examples include the stress-strain relation

$$\mathbf{S} \equiv \mathbf{S}(\boldsymbol{\varepsilon}), \quad \text{in nonlinear elasticity },$$

and the caloric and thermal equations of state in Euler fluids, expressing internal energy and pressure as functions of mass density and temperature.

$$p \equiv p(\rho, \theta), \quad e \equiv e(\rho, \theta).$$

Introducing such equations in the balance laws results in a **hyperbolic differential system**.

- **Non-local in space** - Examples include Fourier's law for heat flux, Navier-Stokes' law for viscous stress, Fick's law for mixture diffusion, and Darcy's law for porous media. Introducing these equations in the balance laws results in a system of differential equations where some spatial derivatives are of second order, and the time derivatives are of first order. These differential systems have then a **parabolic structure**.

- **Non-local in time** - Examples are viscoelastic materials, or any material with memory effect where stress depends not only on present deformation but also on its history. Except for an exponential memory kernel, the mathematical structure of such systems is of **integro-differential type**.

Taking into account the long-standing debate in the literature following Müller seminal work which demonstrated that Fourier and Navier-Stokes "constitutive" equations violate objectivity, I wrote in 2012 a provocative paper titled *Can Constitutive Relations be Represented by Non-local Equations ?* in which basing on considerations of Rational Extended Thermodynamics (RET) it was shown that:

- Navier Stokes' and Fourier's laws are a limit case of hyperbolic balance equations for shear viscous tensor, dynamic pressure, and heat flux in RET (Grad type);
- Fick's law is a limit case of momentum equations of each component in a mixture;
- Darcy's law for porous material is a limit case of the momentum equation when inertial terms are neglected.

Therefore, the conjecture was that "true" physical systems are of hyperbolic type, which in particular is mandatory in the relativistic framework by principle.
All the previous considerations made do not include the case of non-local in-time materials in which the history of the material gives the constitutive equations.
In this talk , I want to present a possibility that also model non-local in time like viscoelasticity can be modeled in such way the differential system is hyperbolic with local constitutive equations. This idea is not new and was also the contents outlined in Chapter 16 of the book by Müller and Ruggeri (1998) and on the Liu's paper (1989).

Viscoelasticity

Viscoelasticity remains an open and fascinating topic, given its broad spectrum of potential applications. Beyond polymeric materials such as rubbers, it encompasses biological tissues like organs, brain, skin, cardiovascular structures, connective tissues, and muscles. All soft materials, characterized by hyperelastic (nonlinear) static behavior, inherently exhibit nonlinear viscoelasticity. The prototype for this behavior in the linear case is the Maxwell model, and various derivatives consider materials with a memory effect, i.e., non-local in time.

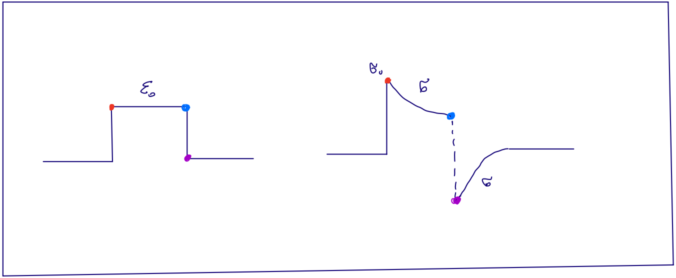
$$\tau\frac{\partial\sigma}{\partial t}-\mu\frac{\partial v}{\partial x}=-\sigma, \quad \Longleftrightarrow \quad \tau\frac{\partial\sigma}{\partial t}-\mu\frac{\partial\varepsilon}{\partial t}=-\sigma.$$

or

$$\sigma=\nu\varepsilon+\int_0^t K(t-t')\dot{\varepsilon}(t')dt' \tag{11}$$

or with fractional derivative:

$$\sigma+\tau_0^\alpha D^\alpha\sigma=bD^\alpha\varepsilon\,, \qquad \alpha\in(0,1)\,. \tag{12}$$


ε_0
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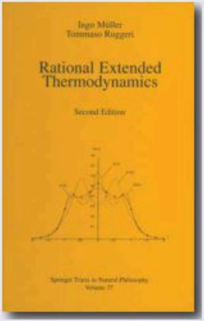
Rational Extended Thermodynamics

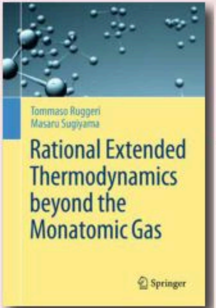
- Rarefied Monatomic Gas

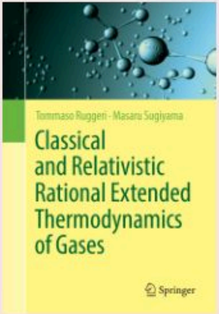
☞ **Müller & Ruggeri, "Rational Extended Thermodynamics" (Springer 1993,1998)**



- Rarefied Polyatomic Gas

☞ **Ruggeri & Sugiyama, (Springer 2015, 2021)**





RET of Rarefied Monoatomic Gas

Boltzmann equation

$$\frac{\partial f}{\partial t} + c_i \frac{\partial f}{\partial x_i} = Q, \qquad F_{i_1 i_2 \cdots i_j} = m \int_{\mathbb{R}^3} f c_{i_1} c_{i_2} \cdots c_{i_j} d\mathbf{c}, \quad \rightarrow$$

$$\partial_t F + \partial_i F_i = 0$$

$$\swarrow$$

$$\partial_t F_{k_1} + \partial_i F_{ik_1} = 0$$

$$\swarrow$$

$$\partial_t F_{k_1 k_2} + \partial_i F_{ik_1 k_2} = P_{<k_1 k_2>}$$

$$\swarrow$$

$$\partial_t F_{k_1 k_2 k_3} + \partial_i F_{ik_1 k_2 k_3} = P_{k_1 k_2 k_3}$$

$$\vdots$$

$$\partial_t F_{k_1 k_2 \ldots k_n} + \partial_i F_{ik_1 k_2 \ldots k_n} = P_{k_1 k_2 \ldots k_n}$$

$$\vdots$$

Viscoelasticity via RET Principles

The goal is to propose a possible new approach to nonlinear viscoelasticity using the RET postulate, i.e. requiring:

- If new fluxes exist, new extended balance laws with local constitutive equations are established;
- The system must be closed, compatible with the universal principles of RET: entropy principle, objectivity, and convexity of the entropy so that the system is symmetric hyperbolic.

For simplicity, we consider the one-space dimension and the isothermal case. We can prove that the universal principles are enough to determine a closed model of viscoelasticity that, in the linear case, reduces to the Maxwell model for what concerns the viscous stress and the Zener model for the total stress.

Non-linear Elasticity

An elastic material in the isothermal and one-dimension case is governed in Lagrangian variables by the following first-order system

$$\begin{aligned} &\rho^* \frac{\partial v}{\partial t} - \frac{\partial T(F)}{\partial X} = \rho^* b, \\ &\frac{\partial F}{\partial t} - \frac{\partial v}{\partial X} = 0, \end{aligned} \tag{13}$$

In the 1-dim, all quantities are scalars and we have

$$F = \frac{\partial x}{\partial X} = 1 + \frac{\partial u}{\partial X}, \quad v = \frac{\partial u}{\partial t} \tag{14}$$

where $u = x(X,t) - X$ is the displacement.
As is well known, if we require that any solutions of (13) are also the solutions of the supplementary energy equation (that has the role of *entropy principle* in the isothermal case)

$$\rho^* \frac{\partial}{\partial t}\left(\frac{v^2}{2} + e(F)\right) - \frac{\partial T(F)v}{\partial X} - \rho^* b v = \mathscr{E} \leqslant 0,$$

we obtain

$$T(F) = \rho^* e_F(F), \qquad \mathscr{E} = 0.$$

Non-linear Viscoelasticity

In the case of viscoelasticity, we assume that the total stress is the sum of the elastic part and the viscous one: $T(F)+\sigma$. Differently, to consider a constitutive equation for σ as typical of hereditary materials, we employ the ideas of RET. Since we have a new flux σ, a new balance law exists to add to (13). Then we consider the following differential extended system of balance laws:

$$\begin{aligned}
&\rho^* \frac{\partial v}{\partial t} - \frac{\partial}{\partial X}(T(F)+\sigma) = \rho^* b, \\
&\frac{\partial F}{\partial t} - \frac{\partial v}{\partial X} = 0, \\
&\frac{\partial \psi(F,\sigma)}{\partial t} + \frac{\partial \Omega(F,\sigma)}{\partial X} = P(F,\sigma).
\end{aligned} \tag{15}$$

We require that any solutions $(v(X,t), F(X,t), \sigma(X,t))$ of the system (15) also satisfy the supplementary energy dissipative balance law:

$$\rho^* \frac{\partial}{\partial t}\left(\frac{v^2}{2} + e(F,\sigma)\right) - \frac{\partial}{\partial X}((T(F)+\sigma)v) - \rho^* b v = \mathscr{E} \leqslant 0. \tag{16}$$

to restrict the local constitutive equations

$$\mathscr{Z} \equiv \{\psi(F,\sigma), \Omega(F,\sigma), P(F,\sigma), e(F,\sigma), \mathscr{E}(F,\sigma)\}.$$

We assume as reasonable that the internal energy is the sum of an elastic part and a viscous one:

$$e(F,\sigma) = e^{(E)}(F) + e^{(V)}(\sigma),$$

we require that the dissipative source $P(F,\sigma)$ vanish with σ:

$$P(F,0) = 0, \tag{17}$$

and

$$\psi_\sigma = \left(\frac{\partial \psi}{\partial \sigma}\right)_F \neq 0, \tag{18}$$

in such a way, the last equation of (15) can be put in normal form with respect to the time derivative of σ:

$$\frac{\partial \sigma}{\partial t} = -\frac{1}{\psi_\sigma}\left\{\psi_F \frac{\partial v}{\partial X} + \Omega_F \frac{\partial F}{\partial X} + \Omega_\sigma \frac{\partial \sigma}{\partial X} - P\right\}. \tag{19}$$

Theorem

A necessary and sufficient condition such that the system (15) *satisfies the dissipative supplementary equation* (16) *is that the constitutive equations satisfy the following expressions:*

$$
\begin{aligned}
& T(F) = \rho^* e_F^{(E)}(F), \\
& \psi(F,\sigma) = \Psi(w), \quad w = F - Z(\sigma), \quad Z(\sigma) = \int \rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma} d\sigma, \\
& \Omega(F,\sigma) = 0, \\
& P(F,\sigma) = \frac{\sigma}{\alpha \Psi'(w)}, \\
& \mathscr{E} = -\alpha P^2, \quad \alpha \equiv \alpha(F,\sigma) > 0,
\end{aligned}
\tag{20}
$$

where Ψ *is an arbitrary function of* w *and* α *is an arbitrary positive function of* (F,σ).

Proof : By eliminating the time derivative in (16) through the equations $(15)_{1,2}$ and from (19) we obtain:

$$\left(\rho^* e_F^{(E)} - \rho^* \frac{e_\sigma^{(V)}}{\psi_\sigma}\psi_F - T - \sigma\right)\frac{\partial v}{\partial X} -$$

$$\rho^* \frac{e_\sigma^{(V)}}{\psi_\sigma}\Omega_F \frac{\partial F}{\partial X} - \rho^* \frac{e_\sigma^{(V)}}{\psi_\sigma}\Omega_\sigma \frac{\partial \sigma}{\partial X} + \rho^* \frac{e_\sigma^{(V)}}{\psi_\sigma} P = \mathscr{E} \leqslant 0.$$

As the inequality needs to be verified for any processes and the spatial derivatives are arbitrary and linear, the only possible solution is

$$\begin{aligned}
&\rho^* e_F^{(E)}(F) - \rho^* \frac{e_\sigma^{(V)}(\sigma)}{\psi_\sigma}\psi_F - T(F) - \sigma = 0,\\
&\Omega_F = \Omega_\sigma = 0, \quad i.e. \quad \Omega = \text{constant} = 0,\\
&\rho^* \frac{e_\sigma^{(V)}}{\psi_\sigma} = -\alpha(F,\sigma)P, \quad \alpha(F,\sigma) > 0,
\end{aligned} \tag{21}$$

and therefore

$$\mathscr{E} = -\alpha P^2 \leqslant 0.$$

Inserting $(21)_3$ into $(21)_1$, we have

$$\rho^* e_F^{(E)}(F) + \alpha P \psi_F - T(F) - \sigma = 0.$$

Putting $\sigma = 0$ and taking into account the assumption (17), we have:

$$T(F) = \rho^* e_F^{(E)}(F),$$

as in the elastic case, while from $(21)_1$ we have the following partial differential equation for $\psi(F,\sigma)$:

$$\psi_\sigma + Z_\sigma(\sigma)\psi_F = 0, \qquad Z_\sigma(\sigma) = \rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma}, \tag{22}$$

that has as a solution

$$\psi(F,\sigma) = \Psi(F - Z(\sigma)),$$

where $\Psi(w)$ is an arbitrary function of $w = F - Z(\sigma)$, with $Z(\sigma)$ given by integration of $(22)_2$. Substituting into $(21)_3$, we have

$$P = \frac{\sigma}{\alpha \Psi'(w)},$$

where the prime indicates the derivative with respect to w. We remark that by the condition (18) $\Psi'(w) \neq 0$.

Substituting the first four expressions of (20), the system (15) becomes equivalent to the following system:

$$\begin{aligned}
&\rho^* \frac{\partial v}{\partial t} - \frac{\partial}{\partial X}(\rho^* e_F^{(E)}(F) + \sigma) = \rho^* b \\
&\frac{\partial F}{\partial t} - \frac{\partial v}{\partial X} = 0, \\
&\frac{\partial}{\partial t}(Z(\sigma) - F) = -\frac{\sigma}{\alpha \Psi'^2}.
\end{aligned} \tag{23}$$

Taking into account the expression of Z given by the $(22)_2$ and the second of (23), the last equation of (23) becomes:

$$\rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma} \frac{\partial \sigma}{\partial t} - \frac{\partial v}{\partial X} = -\frac{\sigma}{\alpha \Psi'^2}, \tag{24}$$

that has the form of a nonlinear Maxwell equation. □

Now, we want to impose the other requirements of RET requiring the convexity condition under which the system is symmetric hyperbolic and the Maxwellian iteration. First, we observe that the equation $(23)_3$ has an infinite form of balance law for the possible choice of the function $\Psi(w)$:

$$\frac{\partial \Psi(w)}{\partial t} = \frac{\sigma}{\alpha \Psi'}, \qquad w = F - Z(\sigma). \tag{25}$$

The system is a particular case of a general quasi-linear system of balance laws:

$$\frac{\partial \mathbf{F}(\mathbf{u})}{\partial t} + \frac{\partial \mathbf{G}(\mathbf{u})}{\partial X} = \mathbf{f}(\mathbf{u}) \tag{26}$$

with

$$\mathbf{F} = \begin{pmatrix} \rho^* v \\ F \\ -\Psi(w) \end{pmatrix}, \qquad \mathbf{G} = \begin{pmatrix} -(T(F)+\sigma) \\ -v \\ 0 \end{pmatrix}, \qquad \mathbf{f} = \begin{pmatrix} \rho^* b \\ 0 \\ -\frac{\sigma}{\alpha\Psi'} \end{pmatrix}, \tag{27}$$

and the energy equation (16) is, in the general case, the supplementary quasi-linear law

$$\frac{\partial h(\mathbf{u})}{\partial t} + \frac{\partial g(\mathbf{u})}{\partial X} = \Sigma(\mathbf{u}), \tag{28}$$

that, in our case, is given by

$$h = \rho^* \left(\frac{v^2}{2} + e^{(E)}(F) + e^{(V)}(\sigma) \right), \quad g = -(T(F)+\sigma)v, \quad \Sigma = \rho^* b v + \mathscr{E}.$$

For this kind of system, we recall that under the thermodynamic stability condition that h is a convex function with respect the field of densities $\mathbf{u} \equiv \mathbf{F}$, Ruggeri and Strumia, starting from the papers by Godunov and Boillat, proved the following theorem.

Theorem (Ruggeri & Strumia)

The compatibility between the system of balance laws (26) *and the supplementary balance law* (28) *with h being a convex function of $\mathbf{u} \equiv \mathbf{F}$ implies the existence of the main field $\mathbf{u}'$ that satisfies*

$$dh = \mathbf{u}' \cdot d\mathbf{F}, \qquad dg = \mathbf{u}' \cdot d\mathbf{G}, \qquad \Sigma = \mathbf{u}' \cdot \mathbf{f}. \tag{29}$$

If we introduce the potentials defined by

$$h' = \mathbf{u}' \cdot \mathbf{F} - h, \qquad g' = \mathbf{u}' \cdot \mathbf{G} - g \tag{30}$$

then $(29)_{1,2}$ *can be rewritten as*

$$dh' = \mathbf{F} \cdot d\mathbf{u}', \qquad dg' = \mathbf{G} \cdot d\mathbf{u}'.$$

Choosing the components of $\mathbf{u}'$ as field variables, we have

$$\mathbf{F} = \frac{\partial h'}{\partial \mathbf{u}'}, \qquad \mathbf{G} = \frac{\partial g'}{\partial \mathbf{u}'},$$

and then the original system (26) *can be rewritten in a symmetric form with Hessian matrices:*

$$\frac{\partial}{\partial t}\left(\frac{\partial h'}{\partial \mathbf{u}'}\right) + \frac{\partial}{\partial X}\left(\frac{\partial g'}{\partial \mathbf{u}}\right) = \mathbf{f} \quad \Longleftrightarrow \quad \frac{\partial^2 h'}{\partial \mathbf{u}' \partial \mathbf{u}'}\frac{\partial \mathbf{u}'}{\partial t} + \frac{\partial^2 g'}{\partial \mathbf{u}' \partial \mathbf{u}'}\frac{\partial \mathbf{u}'}{\partial X} = \mathbf{f}. \tag{31}$$

Theorem

The only system (15) that satisfies the universal principles of RET: Compatibility with a dissipation principle; Convexity, Stability and Symmetric hyperbolic form; Maxwellian Iteration; is the following one:

$$
\begin{aligned}
&\rho^* \frac{\partial v}{\partial t} - \frac{\partial}{\partial X}\left(\rho^* e_F^{(E)}(F) + \sigma\right) = \rho^* b, \\
&\frac{\partial F}{\partial t} - \frac{\partial v}{\partial X} = 0, \\
&\frac{\partial}{\partial t}(Z(\sigma) - F) = -\frac{\sigma}{\mu(F)}, \quad \Longleftrightarrow \quad \rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma} \frac{\partial \sigma}{\partial t} - \frac{\partial v}{\partial X} = -\frac{\sigma}{\mu(F)},
\end{aligned}
\tag{32}
$$

with Z given in $(22)_2$. All constitutive equations are prescribed, and the system is closed, provided we know the internal elastic energy $e^{(E)}(F)$, the internal viscous energy $e^{(V)}(\sigma)$ and the viscosity coefficient $\mu(F)$ that must satisfy the following inequalities:

$$
e_{FF}^{(E)}(F) > 0, \qquad \frac{e_\sigma^{(V)}(\sigma)}{\sigma} > 0, \qquad \mu(F) \geqslant 0. \tag{33}
$$

The system is symmetric hyperbolic choosing as variables the main field

$$
\mathbf{u}' \equiv (v, T(F) + \sigma, \sigma).
$$

Proof - As we have verified the conditions for exploiting the entropy principle with the usual procedure or RET we obtain immediately the following expression:

$$\xi = v, \qquad \zeta = T + \sigma, \qquad \eta = \frac{\sigma}{\Psi'}. \tag{34}$$

The convexity requires that the quadratic form is positive:

$$Q = \frac{\partial^2 h'}{\partial \mathbf{u}' \partial \mathbf{u}'} \delta \mathbf{u}' \cdot \delta \mathbf{u}' = \delta \left(\frac{\partial h'}{\partial \mathbf{u}'} \right) \cdot \delta \mathbf{u}' = \delta \mathbf{F} \cdot \delta \mathbf{u}' > 0.$$

Taking into account (34) and (27), it is simple to verify that necessary and sufficient condition such that $Q > 0$ for any processes is:

$$e^{(E)}_{FF}(F) > 0, \qquad \frac{e^{(V)}_{\sigma}(\sigma)}{\sigma} > 0, \qquad \Psi''(w) = 0. \tag{35}$$

The first inequality is the usual convexity condition for nonlinear elasticity; the second one if we want that the expression is bounded also for $\sigma \to 0$ implies in particular:

$$e^{(V)}_{\sigma}(0) = 0, \qquad e^{(V)}_{\sigma\sigma}(0) > 0, \tag{36}$$

while the last condition requires that Ψ' is constant and without loss of generality, we can put equal to 1 taking into account that the constant can incorporate in α in equation (24) and therefore $\Psi = F - Z(\sigma)$.

Now, as usual in RET, the parabolic limit of the system can be obtained using the *Maxwellian Iteration* proposed first by Ikenberry and Truesdell. In the present case, when we neglect the time derivative in (24), we need to reduce to Navier-Stokes equation with viscosity coefficient $\bar{\mu}$ that in general is a function of F:

$$\sigma = \bar{\mu}(F)\,\frac{\partial v}{\partial x} = \mu(F)\,\frac{\partial v}{\partial X}, \qquad \text{with} \qquad \mu(F) = \frac{\bar{\mu}(F)}{F},$$

and therefore it is possible to identify $\alpha = \mu$ and according with the last inequality is (20) we have $\mu(F) \geqslant 0$, and the theorem is proved. $\square$

In the nonlinear case, the deformation tensor in $1-$dim is

$$\varepsilon = \frac{\partial u}{\partial X} + \frac{1}{2}\left(\frac{\partial u}{\partial X}\right)^2,$$

and therefore from (14) we have

$$F = \sqrt{1+2\varepsilon}. \tag{37}$$

Inserting in the last equation of (32), we have the alternative form

$$\rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma}\frac{\partial \sigma}{\partial t} - \frac{1}{\sqrt{1+2\varepsilon}}\frac{\partial \varepsilon}{\partial t} = -\frac{\sigma}{\mu(\varepsilon)}. \tag{38}$$

We need to observe that σ is only the viscous part of the stress. If we wan to rewrite the equation (38) in terms of the total stress:

$$S = T(F) + \sigma = \rho^* e_F^{(E)}(F) + \sigma \tag{39}$$

we have:

$$\rho^* \frac{e_\sigma^{(V)}(\sigma)}{\sigma}\frac{\partial S}{\partial t} - \frac{1}{\sqrt{1+2\varepsilon}}\left(\rho^{*2}\frac{e_\sigma^{(V)}(\sigma)}{\sigma} e_{FF}^{(E)}(F) + 1\right)\frac{\partial \varepsilon}{\partial t} = -\frac{S}{\mu(F)} + \rho^* \frac{e_F^{(E)}(F)}{\mu(F)} \tag{40}$$

In the linear case $F=1+\varepsilon$, where $\varepsilon=\frac{\partial u}{\partial x}$ is the deformation tensor in 1-dim, $\mu=$ const. and $x=X$. Choosing the energy quadratic:

$$e^{(E)}=\frac{\nu}{2\rho^*}\varepsilon^2,\quad e^{(V)}=\frac{\tau}{2\rho^*\mu}\sigma^2,$$

with ν is the Young modulus and the constant τ is a relaxation time, the system (32) becomes linear with the Hooke law for the elastic part $T=\nu\,\varepsilon$ and the last equation become the Maxwell equation of linear viscoelasticity for what concerns the viscous stress:

$$\tau\frac{\partial\sigma}{\partial t}-\mu\frac{\partial v}{\partial x}=-\sigma,\quad\Longleftrightarrow\quad\tau\frac{\partial\sigma}{\partial t}-\mu\frac{\partial\varepsilon}{\partial t}=-\sigma.$$

Therefore, the equation (38) is a natural nonlinear extension of the Maxwell case. Concerning the total stress given in (40) this reduces to

$$\tau\frac{\partial S}{\partial t}-(\nu\tau+\mu)\frac{\partial\varepsilon}{\partial t}=-S+\nu\,\varepsilon\tag{41}$$

that is a *standard linear solid model* in the Maxwell representation. Therefore, we can consider (38) as the nonlinear Maxwell model for the viscous stress, and for what concerns the total stress, the equation (40) is a nonlinear variant of the Zener model.

It is essential to notice that these equations come without any direct postulations by only the universal principles of Rational Extended Thermodynamics.

Physical interpretation

In this simple case of 1-dimension and for classical solutions, the new balance law $(32)_3$ has a simple physical interpretation. In fact, if we multiply it by σ, we can rewrite this equation as:

$$\rho^* \frac{\partial e^{(V)}(\sigma)}{\partial t} = \sigma \frac{\partial v}{\partial X} - \frac{\sigma^2}{\mu}. \tag{42}$$

This equation represents the time evolution of the viscous energy, which is partly transformed into the power of the viscous stress, and part is lost due to dissipation. It is important to note that this consequence arises from universal principles in one space dimension, as the new balance law $(15)_3$ is, in this circumstance, a scalar equation. In three dimensions of space, the viscous stress is a tensor denoted as σ_{iA}, leading to the new balance $(15)_3$ becoming a tensorial equation. Therefore, the 3-dimensional case requires further investigation.

Characteristic velocities

We have the following algebraic homogeneous characteristic system:

$$\begin{aligned}
&\lambda\,\rho^*\delta v + \rho^* e_{FF}^{(E)}(F)\,\delta F + \delta\sigma = 0,\\
&\lambda\,\delta F + \delta v = 0,\\
&\lambda\,\rho^*\frac{e_\sigma^{(V)}(\sigma)}{\sigma}\delta\sigma + \delta v = 0.
\end{aligned} \tag{43}$$

It is simple to verify that the solutions of (43) are:

$$\lambda = 0 \quad \text{(contact wave)}, \quad \delta v = 0, \quad \delta\sigma = -\rho^* e_{FF}^{(E)}(F)\,\delta F, \quad \delta F \neq 0, \tag{44}$$

and the sound velocities:

$$\lambda = \pm\sqrt{e_{FF}^{(E)} + \frac{\sigma}{\rho^{*2} e_\sigma^{(V)}}}, \quad \text{with} \quad \delta\sigma = -\frac{\sigma}{\lambda\,\rho^* e_\sigma^{(V)}(\sigma)}\delta v, \quad \delta F = -\frac{1}{\lambda}\delta v, \quad \delta v \neq 0. \tag{45}$$

Taking into account the inequalities (35), the characteristic velocities are real and the right eigenvectors are linearly independent; in agreement with the fact that every symmetric system is automatically hyperbolic.

Principal subsystem and subcharacteristic conditions

Moreover, the system of elasticity (13) is a *principal subsystem* of the new one (32) when $\sigma = 0$ according to the definition and properties given in Boillat-Ruggeri (ARMA 1997). Then, it automatically satisfies the subcharacteristic condition with respect to the elastic case, as is also evident from (45) and (36):

$$\lim_{\sigma\to 0}\left(e^{(E)}_{FF} + \frac{\sigma}{\rho^{*2} e^{(V)}_{\sigma}}\right) = e^{(E)}_{FF} + \frac{1}{\rho^{*2} e^{(V)}_{\sigma\sigma}(0)} > e^{(E)}_{FF} > 0.$$

Therefore the sound velocity in viscoelasticity is greater concerning the ones of the elastic case.

K-condition and global solutions

Moreover, the so-called K-condition, applied to the system (26), is given by the requirement that in the characteristic algebraic system $\delta \mathbf{f}|_{eq} \neq 0$ (where *eq* denotes equilibrium). In this context, it is equivalent to the condition $\delta \sigma|_{\sigma=0} \neq 0$. Notably, from the present characteristic system (43), this condition is satisfied for both the contact wave (see (44)) and for the sound waves, where from $(45)_2$, we have

$$\delta \sigma|_{\sigma=0} = -\frac{1}{\lambda\, \rho^* e^{(V)}_{\sigma\sigma}(0)} \delta v \neq 0,$$

for the inequality $(36)_2$.

Consequently, the differential system exhibits globally smooth solutions for small initial data, aligning with well-established theorems (Yong, Bianchini, Hanounzet & Natalini, Ruggeri & Serre).

Conclusions

We systematically formulate a hyperbolic model for nonlinear viscoelasticity in one space dimension, employing the universal principles of Rational Extended Thermodynamics. The resulting differential system is symmetric hyperbolic, satisfies an entropy principle with a convex entropy, and adheres to the K-condition. Consequently, global smooth solutions exist for suitable initial data. The equation governing the viscous stress, $(32)_3$, or equivalently (38), naturally extends for the viscous stress the linear Maxwell equation for viscoelasticity and represents the decay in time of the viscous energy, as given by (42). Moreover, for what concerns the total stress, the present model indicates a nonlinear version of the so-called Zener model.

The contents of this presentation is in my recent paper:
T. Ruggeri, **A nonlinear approach to viscoelasticity via rational extended thermodynamics**, Int. J. Non-Linear Mech. 160, 104658 (2024).

Recent work

In a recent work jontly with A. Giusti and A. Mentrelli "*Energy of a non-linear viscoelastic model compatible with fractional relaxation*" we investigate under which conditions the relaxation modulus of the linear Fractional Maxwell model, defined by the relation

$$\sigma + \tau_0^\alpha D^\alpha \sigma = b D^\alpha \varepsilon \,, \qquad \alpha \in (0,1)\,, \tag{46}$$

where τ_0 and b are real dimensional constants and D^α denoting the Caputo fractional derivative with respect to time, can be reproduced by RET-improved viscoelasticity and relying on the local non-linear equation in (38).
The *relaxation modulus* is defined as the solution of the stress relaxation experiment or, in other words, it is the solution of the constitutive equation for a material given a constant strain $\varepsilon(t) = \varepsilon_0$. Specifically, the relaxation modulus for the local non-linear model in Eq. (38) is the solution of

$$\sigma + \rho^* \mu(\varepsilon_0) \frac{e_\sigma^{(V)}(\sigma)}{\sigma} \, \dot{\sigma} = 0\,. \tag{47}$$

For the linear fractional Maxwell model of order $\alpha \in (0,1)$, the relaxation modulus is a solution of

$$\sigma + \tau_0^\alpha D^\alpha \sigma = 0\,, \qquad \alpha \in (0,1)\,. \tag{48}$$

Theorem (A. Giusti, A. Mentrelli & T. Ruggeri 2024)

Consider the family of constitutive equations $e^{(V)}(\sigma)$ *depending on* α *and expressed in the following parametric form:*

$$
\begin{aligned}
e^{(V)}(s) &= e_0 - \frac{k_0^2}{\rho^* \mu(\varepsilon_0)} \int_0^s \Big(E_\alpha \big[-(\bar{s}/\tau_0)^\alpha \big] \Big)^2 d\bar{s}, \\
\sigma(s) &= k_0 \, E_\alpha \big[-(s/\tau_0)^\alpha \big],
\end{aligned} \tag{49}
$$

with $s \geqslant 0$, e_0 *being a real inessential constant, and* k_0 *a structural constant of the material. Then, there exists a solution* $\sigma(t) = k_0 \, E_\alpha \big[-(t/\tau_0)^\alpha \big]$ *of both Eqs.* (47) *and* (48) *if we choose as initial condition* $\sigma(0) = k_0$. *The viscous energy* $e^{(V)}(\sigma)$ *is bounded* $\forall \sigma \in [0, k_0]$ *if and only if* $E_\alpha[-(s/\tau_0)^\alpha] \in L^2\big([0,\infty)\big)$, *namely, if and only if* $\alpha \in (1/2, 1)$. *The non-linear relaxation time computed assuming* $\varepsilon(t) = \varepsilon_0$, *is positive for all* $\sigma \in (0, k_0)$, *diverges at* $\sigma = 0$, *and vanishes at* $\sigma = k_0$.

Work in Progress

- Extension to non-isothermal case - T. Arima & T. Ruggeri .
- Creep and Stress Relaxation - T. Arima & T. Ruggeri .
- Shock waves and existence of continuous solutions - T. Arima, T. Ruggeri, & S. Taniguchi.